Elementary Number Theory

(Notes by Tat Wing Leung 22, 29 Sept., 2018)

N = {1, 2, 3, …}, natural numbers
Z = {…, -3, -2, -1, 0, 1, 2, 3, …}, integers, also $Z^{\geq 0}$ or $Z^{>0}$.
Q = {p/q, p, $q \in Z$, $q \neq 0$}, rational numbers
R = real numbers
C = complex numbers

Divisibility

1. If $b = aq$ for some $q \in \mathbf{Z}$, then a divides b, (or a a factor of b, or b divisible by a), write $a \mid b$.

2. Divisibility relations:
   I. $a \mid b$, $b \mid c$, then $a \mid c$.
   II. $d \mid a$, $d \mid b$, then $d \mid ax + yb$, especially $d \mid a + b$, and $d \mid a - b$.
   III. If any two terms in $a + b = c$ are divisible by d, so is the third term.

3. Divisibility with remainder. Every integer a and positive integer b, there is the unique representation $a = bq + r$, with $0 \leq r < b$. q and r are called quotient and remainder upon division of a by b.

4. GCD and Euclidean Algorithm: Let a and b be non-negative integers, not both 0. The greatest common divisor of a and b (denoted by gcd(a, b)) is the largest positive integer that divides both a and b, the least common multiple (denoted by lcm (a, b)) of a and b is the smallest positive integer divisible by both a and b. We observe a few trivial cases:
   gcd(a, 1) = 1, gcd (a, a) = a, gcd (a, ac) = a, gcd (a, b) = gcd (b, a).
   a and b are relatively prime if gcd (a, b) = 1.
   Note if $b = aq + r$, then gcd (a, b) = gcd (b, r), by writing $b = rq_1 + r_1$, and proceed in the same manner, we find the gcd (a, b) eventually (Euclidean algorithm).

   Example: $a = 187$, $b = 35$;
   $187 = 5 \times 35 + 12$, $35 = 2 \times 12 + 11$, $12 = 1 \times 11 + 1$, $11 = 11 \times 1$, obtain gcd (187, 35) = 1.
   Also $1 = 12 - 11 \times 1 = 12 - (35 - 2 \times 12) = 3 \times 12 - 35 = 3 \times (187 - 5 \times 35) - 35 = 3 \times 187 - 16 \times 35$, i.e., 1 is an integer linear combination of 187 and 35.

   In general: gcd (a, b) can be represented as a linear combination of the form $ax + by$.

   Moreover denote the set $S = \{ax + by \mid x, y \in \mathbf{Z}\}$. Then gcd ($a$, b) = smallest positive element in S. For if d is such member, and if $e \mid a$, $e \mid b$, then $e \mid d$. Also for any member $u \in S$, then $u = qd + r$, $0 \leq r < d$, hence $r = u - qd$. But note that in this case $r \in S$, by minimality of d, get $r = 0$. Hence $d \mid a$ and $d \mid b$.

5. gcd (a, b) lcm(a, b) = ab.

Prime Numbers

1. A positive integer p (>0) is <u>prime</u> if its only positive factors are 1 and itself. The primes are 2, 3, 5, 7, 11,…. Now $n > 1$, if it is not a prime, then $n = ab$, by induction, n is a product of primes.

2. If p is a prime, $p \mid ab$, then $p \mid a$ or $p \mid b$.

3. <u>Fundamental Theorem of Arithmetic</u>: Every integer can be represented as a product of primes, representation is unique up to rearrangement. (Proof by induction).

4. <u>Theorem</u>: There exist infinitely many primes. (Consider $p_1 p_2 \cdots p_n + 1$).

5. $n! + 2$, $n! + 3$, …, $n! + n$ are $n - 1$ consecutive composite (nonprime) numbers. Using this, may show there exists a sequence of 2018 consecutive integers that contains exactly one prime. Just go up from such a sequence until the first prime. Then go down.

6. The smallest prime factor of a nonprime n is $\le \sqrt{n}$.

7. All pairwise prime (primitive) triplets of integers satisfying $x^2 + y^2 = z^2$ are given by

   $x = |u^2 - v^2|$, $y = 2uv, z = u^2 + v^2, \gcd(u, v) = 1, u - v \neq 2n$. (Also: consider the straight lines from $(-1, 0)$ and go through the unit circle $x^2 + y^2 = 1$ may get all rational solutions and hence the integer solutions of the equation $x^2 + y^2 = z^2$ (parametrization)).

<u>Exercises 1</u>

1. Find all integer solutions of $x^2 - y^2 = 2018$.
2. Let a, b, c and d be natural numbers, and $a + b + c + d = 2018$, is $a^2 + b^2 + c^2 + d^2$ even or odd?
3. Show that the product of two integers of the form $4n + 3$ is also an integer of that form. Hence show that there exist infinitely many primes of the form $4n + 3$.
4. Let each of $a_1, a_2, \cdots, a_n$ be either 1 or –1. If

   $a_1a_2a_3a_4 + a_2a_3a_4a_5 + \cdots + a_na_1a_2a_3 = 0$, show that $4 \mid n$.

   (First n is even, then consider the product of all the terms. Or change −1 to 1 using invariant).
5. If a is not divisible by 2 or 3, show that $a^2 - 1$ is a multiple of 24.
6. Find gcd(329, 182).
7. (IMO1959, 1) Show that the fraction $\dfrac{21n+4}{14n+3}$ is irreducible.
8. For $n > 1$, $1 + \dfrac{1}{2} + \cdots + \dfrac{1}{n}$ is not an integer. (If $2^s \le n < 2^{s+1}$, multiply the expression by $2^{s-1}Q$, where Q is the product of all odd factors from 1 to n).
8. Show that 641 divides $2^{32} + 1$ (without using a calculator).
9. Suppose p is prime, then p divides C_r^p, with $1 \le r \le p - 1$.

<u>Basic Prime Number Facts</u>

1. Denote by $\pi(x)$ the number of primes less than or equal to x, then $\pi(x) \sim \frac{x}{\log x}$, (Prime number theorem).
2. (Euler): $\sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \frac{1}{(1-\frac{1}{p^s})}, s > 1$. Taken limit as s goes to 1, there exist infinitely primes.

   But in fact $\sum \frac{1}{p}$ diverges.
3. $2^m + 1$ prime, then $m = 2^n$. Fermat claimed all numbers of the form $F_n = 2^{2^n} + 1$ is prime. True for $n = 0, 1, 2, 3, 4$. But 641 divides F_5. In fact:
   $2^{32} = 2^4 2^{28} = (641 - 5^4)2^{28} = 641m - (5 \cdot 2^7)^4 = 641m - (641-1)^4 = 641n - 1.$
4. $2^n - 1$ prime, then n is prime (Mersenne prime).
5. There is a prime lying between n and $2n$ (Bertrand's Postulate).
6. There exist infinitely many prime of the form $a + nb$, if gcd(a, b) =1 (Dirichlet).

Congruence

a congruent to b modulo n if $n | a - b$. Write $a \equiv b \pmod{n}$.

If $0 \le b < n$, and $a \equiv b \pmod{n}$, then b the residue of a mod n. 0, 1, 2,…, n-1, the complete set of residues mod n.

Facts:

(i) $a \equiv b \pmod{n}$ is an equivalence relation.

(ii) $a \equiv b, c \equiv d \pmod{n} \Rightarrow a \pm c \equiv b \pm d \pmod{n}, ac \equiv bd \pmod{n}$.

(iii) $ka \equiv kb \pmod{n}$, gcd(k,n) = 1, then $a \equiv b \pmod{n}$. Thus if $a_1, a_2, \cdots, a_n$ complete set of residues mod n, so is $ka_1, ka_2, \cdots, ka_n$.

(iv) $ax \equiv b \pmod{n}$ solvable iff gcd (a,n) divides b. Example gcd (a, n) = 1, or n prime and gcd (a, n) = 1, gets a unique solution.

Chinese Remainder Theorem: $n_1, n_2, \cdots, n_k$ relatively prime integers, $a_1, a_2, \cdots, a_k$ integers. The system: $x \equiv a_1 \pmod{n_1}, x \equiv a_2 \pmod{n_2}, \cdots, x \equiv a_k \pmod{n_k}$ has a unique solution mod $n_1 n_2 \cdots n_k$.

Proof: Let $n = n_1 n_2 \cdots n_k$, $m_i = \frac{n}{n_i}$ then gcd $(m_i, n_i) = 1$. There exists x_i such that $m_i x_i \equiv a_i \pmod{n_i}$.

Take $x = m_1 x_1 + m_2 x_2 + \cdots + m_k x_k$, then $x \equiv a_i \pmod{n_i}$. Also $x \equiv y \pmod{n_i}$ implies $x \equiv y \pmod{n}$. (Uniqueness).

Example (IMO1989): Construct n consecutive positive integers such that none is a power of prime.
Try $x \equiv -1 \pmod{p_1 p_2}, x \equiv -2 \pmod{p_3 p_4}, \cdots, x \equiv -n \pmod{p_{2n-1} p_{2n}}$.

Reduced set of residue mod n: Elements a of the from $0 \le a < n$, gcd $(a, n) = 1$, The number of such reduced residues = $\phi(n)$.

Fact: ϕ is multiplicative, i.e., if gcd (m, n) = 1, then $\phi(mn) = \phi(m)\phi(n)$.

Proof: $\{a\}$, reduced set of residues mod m, $\phi(m)$, $\{b\}$, reduced set of residues mod n, $\phi(n)$. Consider numbers of the form $an + bm$. Observe (i) gcd ($an + bm$, m) = gcd ($an + bm$, n) = 1. (ii) $an + bm = a'n + b'm$ (mod mn) then $a = a'$ (mod m) and $b = b'$ (mod n). (iii) gcd (c, mn) = 1, gcd (m, n) = 1, then $mt + ns = 1$, then gcd (m, s) = gcd (n, t) = 1. Thus gcd (cs, m) = 1, then $cs = a$ (mod m), gcd (ct, n) = 1, then $ct = b$ (mod n). And $c = cns = an$ (mod m), $c = ctm = bm$ (mod n), then $c = an + bm$ (mod mn).

Fact: If p is prime, $\phi(p^m) = p^m(1-\frac{1}{p})$.

Generally speaking, if $n = p_1^{n_1} p_2^{n_2} ... p_n^{n_2}$, where the p_i's are distinct primes, then $\phi(n) = n\prod(1-\frac{1}{p})$, this can also be proved using the exclusion-inclusion principle ($\phi(n) = n - \sum\frac{n}{p_i} + \sum\frac{n}{p_i p_j} - ... = n\prod(1-\frac{1}{p_i})$, where p_i's are the distinct primes of n.

Now if $x_1, x_2, \cdots, x_{\phi(n)}$ a complete set of reduced residues mod n, so is $ax_1, ax_2, \cdots, ax_{\phi(n)}$, if gcd ($a$, n) = 1, get $a^{\phi(n)} \equiv 1 (\text{mod}\, n)$ (Euler's theorem). $n = p$ prime, $a^{p-1} \equiv 1 (\text{mod}\, p)$ (Fermat's Little Theorem). This can also be proved by induction. Or a combinatorial proof: Consider pearls on a colors, take p of them to form a necklace, a^p possible choices, minus those with only 1 color, get $a^p - a$ choices. The necklace so formed will be indistinguishable after a cyclic permutation of order p, or p divides $a^p - a$.

Let p be prime, for any a between 1 and $p - 1$, there exists a unique b such that $ab = 1$ (mod p). Also $a^2 = 1$ (mod p) iff $a = 1, -1$ (mod p). Hence $2.3\ldots.(p-2)=1$ (mod p) or $1.2.3\ldots(p-1)=-1$ (mod p) (Wilson's theorem.)

Application: $p \equiv 1$ (mod 4), then $-1 = (1.2\ldots.2n)((2n+1)\ldots(4n)) = (1.2\ldots.2n)((-2n)(-(2n-1))\ldots(-1)) = (1.2\ldots2n)^2$, then $x^2 + 1 \equiv 0$ (mod p) solvable. On the other hand, $x^2 \equiv -1$ not solvable if $p \equiv 3 (\text{mod}\, 4)$. Otherwise $x^2 \equiv -1$ (mod p), $x^{p-1} \equiv x^{4n+2} \equiv (x^2)^{2n+1} \equiv (-1)^{2n+1} \equiv -1$ (mod p), contradicting Fermat's little theorem.

Example: Consider $2^p - 1$, we want to find if there is any q prime, that divides $2^p - 1$, this means $2^p \equiv 1 (\text{mod}\, q)$. On the other hand, by the Fermat little theorem $2^{q-1} \equiv 1 (\text{mod}\, q)$. If $d = \gcd(p, q-1)$, then d is of the form $sp + t(q-1)$, hence $2^d = 2^{sp+t(q-1)} = (2^p)^s (2^{q-1})^t \equiv 1 (\text{mod}\, q)$. As d is either 1 or p, it is not 1, otherwise $2 \equiv 1 (\text{mod}\, q)$, which is absurd, hence $d = p$, and p must divide $q-1$, or q is of the form $kp+1$. Also because the number is odd, k is even.

Try $2^{11} - 1$, as $p = 11$, q is of the form $11k + 1$, with k even. Hence $q = 23, 67, 89$, etc. In fact $2^{11} - 1 = 23 \times 89$. Try $2^{13} - 1$, as $p = 13$, $q = 53, 79$, etc. But none of them work, hence $2^{13} - 1$ is prime. Try $2^{23} - 1$. As $p = 23$, $q = 47, 93$, etc. In fact $2^{23} - 1 = 8388607 = 47 \times 178481$.

Example (IMO 1984) Find one pair of positive integers a, b such that (1) $ab(a+b)$ is not divisible by 7, (2) $(a+b)^7 - a^7 - b^7$ is divisible by 7^7.

$(a+b)^7-(a^7+b^7)=7ab(a+b)(a^2+ab+b^2)^2$. Hence it suffices to choose a, b so that $7^3\,|(a^2+ab+b^2)$ or $a^3\equiv b^3 \pmod{7^3}$. By Euler $c^{\phi(7^3)}=c^{7^3-7^2}=c^{3\times 98}\equiv 1 \pmod{7^3}$. We can choose $b=1$ and $a=2^{98}$ to get an answer. Moreover $2^{98}\equiv 4 \pmod 7$, $a+b\equiv 5 \pmod 7$ and $a-b\equiv 3 \pmod 7$, thus conditions satisfied. On can check $2^{10}=3\times 7^3-5\equiv -5 \pmod{7^3}$ and after some work $2^{98}\equiv 18 \pmod{7^3}$ to get the pair (18, 1).

Example (USAMO 1991): Fix a positive integer $n\geq 1$, show that the sequence $2, 2^2, 2^{2^2}, 2^{2^{2^2}}, \ldots \pmod n$ becomes constant eventually.

Proof by induction. True for $n=1,2$, say. Assume the statement is true for all moduli less than or equal to $n-1$. Now let $n=2^k q$, where q is odd and $k\geq 1$. By induction, the sequence $a_1=2, a_2=2^{a_1},\ldots,a_i,2^{a_i},\ldots$ becomes constant mod q eventually. Likewise when i large enough, $a_i\equiv 0 \pmod{2^k}$. Now since 2^k and q are relatively prime, from $2^k\,|a_{i+1}-a_i$ and $q\,|a_{i+1}-a_i$, get $n\,|a_{i+1}-a_i$, when i large enough. When q is odd, then there exist $r<q$ such that $2^r\equiv 1 \pmod n$, for instance may let $r=\phi(n)$. Hence $a_i\equiv c \pmod r$ when i large enough. This implies $a_{i+1}=2^{a_i}=2^{m_i r+c}=(2^r)^{m_i}2^c\equiv 2^c \pmod n$ and the result.

Example (IMO1988): a and b are positive integers satisfying $(1+ab)\,|(a^2+b^2)$, show that $\dfrac{a^2+b^2}{1+ab}$ is a perfect square.

By contradiction, there exists an integer $\dfrac{a^2+b^2}{1+ab}$ which is not a perfect square. Choose a,b such that $\max(a,b)$ is smallest possible. Let $k=\dfrac{a^2+b^2}{1+ab}$. Clearly $a\neq b$, otherwise $k<2$ or $k=1$ (a perfect square). We may then assume $a>b$. Note $a^2+b^2-k(1+ab)=a^2-abk+(b^2-k)=0$ is a quadratic equation of a, whose sum of roots are kb and product b^2-k. Therefore there is another root a' satisfying $a+a'=kb$ and $aa'=b^2-k$. Now $(a')^2+b^2-k(1+a'b)=0$, and as k and b' positive integer, we must have $a'\geq 0$. Indeed $a'>0$ since k is not a perfect square. This forces $a'<\dfrac{b^2-k}{a}<\dfrac{b^2-k}{b}<b$, or $\max(a',b)=b<a=\max(a,b)$, contradicting the minimality of $\max(a,b)$. (This is the method of *infinite descent* by Fermat.)

Now if a is relatively prime to n, the smallest positive integer d such that $a^d\equiv 1 \pmod n$ is the order of a ($\mathrm{ord}_n(a)$ or $\mathrm{ord}(a)$. If $a^{\phi(n)}\equiv 1 \pmod n$, but $a^s\neq 1 \pmod n$ for smaller s, then a is a primitive root mod n.

Fact: Let a be relatively prime to m, then $a^n\equiv 1 \pmod m$ if and only if $ord(a)\,|\,n$.
Furthermore $a^{n_0}\equiv a^{n_1}$ if and only if $\mathrm{ord}(a)\,|(n_0-n_1)$.
Remark: Using Euler's theorem, $\mathrm{ord}(a)\,|\,\phi(m)$.
Example: $\mathrm{ord}_{101}(2)=100$.

Indeed $\phi(101)=100$ (Fermat's little theorem), hence $\text{ord}_{101}(2)\big|100$. If the order of 2 is not 100, then it divides 20 or 50. But check that $2^{50}\equiv -1(\text{mod}\,101)$ and $2^{20}\equiv -6(\text{mod}\,101)$.

Example: If p is a prime number, then every prime divisor of 2^p-1 is greater than p.

If q is a prime divisor of 2^p-1, then $2^p\equiv 1(\text{mod}\,q)$. Hence $\text{ord}_q(2)\big|p$, or the order of 2 must be 1 or p. Clearly it is not 1, it must be p. On the other hand, $2^{q-1}\equiv 1(\text{mod}\,q)$ (Fermat's little theorem), thus $p\big|(q-1)$, or $q>p$. (Indeed if $p>2$, i.e., p is an odd prime, then $q=mp+1$ is odd, so m must be even. (Primes of the form 2^p-1 are called Mersenne primes.)

Example: Let p be a prime and $(p,10)=1$. Let n be an integer such that $0<n<p$. Let d be the order of 10 modulo p.

(a) The length of period of the decimal expansion of $\dfrac{n}{p}$ is d.

(b) If d is even, then the period of the decimal expansion of $\dfrac{n}{p}$ can be divided into two halves, whose sum is $10^{\frac{d}{2}}-1$. (For example, $\dfrac{1}{7}=0.\overline{142857}$, so $d=6$, and $142+857=999=10^3-1$.)

(a) Indeed if $\dfrac{n}{p}=0.\overline{a_1a_2...a_m}$, ($m$ is the length of the period, then $\dfrac{(10^m-1)n}{p}=\overline{a_1a_2...a_m}$ is an integer, hence $p\big|10^m-1$ (p and n relatively prime), so d divides m. Conversely p divides 10^d-1, so $\dfrac{(10^d-1)n}{p}$ is an integer with at most d digits. Dividing the integer by 10^d-1, we get a rational number with decimal expansion of period at most d. Thus $m=d$.

(b) Let $d=2k$, then $\dfrac{n}{p}=0.\overline{a_1...a_ka_{k+1}...a_{2k}}$. Now d divides $10^{2k}-1=(10^k+1)(10^k-1)$, but it does not divide 10^k-1, since the order of 10 is $2k$, so p divides 10^k+1. Hence
$\dfrac{10^k n}{p}=a_1a_2...a_k.\overline{a_{k+1}...a_{2k}a_1a_2...a_k}$, or
$\dfrac{(10^k+1)n}{p}=a_1a_2...a_k+0.\overline{a_1...a_k}+0.\overline{a_{k+1}...a_{2k}}$ is an integer. This implies
$a_1a_2...a_k+a_{k+1}a_{k+2}..a_{2k}$ consisting only of 9s, done.

Example: $n>1$. Show that $n\big|\phi(2^n-1)$.

Indeed $2^n\equiv 1(\text{mod}\,2^n-1)$. And if $1\le k<n$, then $2\le 2^k<2^n$, so $2^k\not\equiv 1(\text{mod}\,2^n-1)$. This implies 2 is of order n $\text{mod}\,2^n-1$. It implies $n\big|\phi(2^n-1)$. (Note we may simply replace 2 by any number a.

Fact: $2^{2\cdot 3^{n-1}}\equiv 1+3^n \quad (\text{mod}\,3^{n+1})$, for all $n\ge 1$.

Assume the statement is true for $n=k$. Then $2^{2\cdot 3^{k-1}}=1+3^k+3^{k+1}m$. Hence by cubing, have $2^{2\cdot 3^k}=1+3^{k+1}+3^{k+2}M$, thus $2^{2\cdot 3^k}\equiv 1+3^{k+1} \pmod{3^{k+2}}$.
Using this, we get

Fact: 2 is a primitive root modulo 3^n for all $n\geq 1$.

The statement is true for $n=1$. We assume the statement is true for $n=k$, thus $2^{\phi(3^k)}\equiv 2^{2\cdot 3^{k-1}}\equiv 1 (\bmod 3^k)$. If d is the order of 2 modulo 3^{k+1}, then $2^d\equiv 1(\bmod 3^{k+1})$, hence $2^d\equiv 1(\bmod 3^k)$, or $2\cdot 3^{k-1}\,|\,d\cdot$ However $d\,|\,\phi(3^{k+1})=2\cdot 3^k$. This implies d is either $2\cdot 3^{k-1}$ or $2\cdot 3^k$. From the previous fact, $2^{2\cdot 3^{k-1}}\equiv 1+3^k\not\equiv 1 \pmod{3^{k+1}}$. Hence $d=2\cdot 3^k$.

Example: If $2^n\equiv -1 (\bmod 3^k)$, then $3^{k-1}\,|\,n$.
Indeed $\phi(3^k)=3^k-3^{k-1}=2\cdot 3^{k-1}$. Also $2^{2n}\equiv 1 (\bmod 3^k)$, hence $2\cdot 3^{k-1}\,|\,2n$, or $3^{k-1}\,|\,n$.
Fact: If m has a primitive root, then m has $\phi(\phi(m))$ primitive roots.
There are $\phi(m)$ invertible elements modulo m. If there is a primitive root g, then so are g^k if $(k,\phi(m))=1$ and there are $\phi(\phi(m))$ of them.
Fact: If g is a primitive root of m, then $g^n\equiv 1(\bmod m)$ iff $\phi(m)\,|\,n$. Moreover $g^{n_0}\equiv g^{n_1} (\bmod m)$ iff $\phi(m)\,|\,n_0-n_1$.
Fact: If g is a primitive root of m, then the powers $1,g,g^2,\ldots,g^{\phi(m)-1}$ represent each integer relatively prime to m uniquely modulo m. In particular if $m>2$, then $g^{\phi(m)/2}\equiv -1 (\bmod m)$.

Using the above fact, if g is a primitive root and a an integer relatively prime to m, then there exist a unique integer i modulo $\phi(m)$ such that $g^i\equiv a (\bmod m)$. The i is the *index* of a with respect to the base g, denoted by $\operatorname{ind}_g(a)$, (the index depends on the choice of g). We have
(1) $\operatorname{ind}_g(1)=0 \pmod{\phi(m)}$, $\operatorname{ind}_g(g)=1 \pmod{\phi(m)}$,
(2) $a\equiv b (\bmod m)$, then $\operatorname{ind}_g(a)=\operatorname{ind}_g(b) \pmod{\phi(m)}$,
(3) $\operatorname{ind}_g(ab)=\operatorname{ind}_g(a)+\operatorname{ind}_g(b) \pmod{\phi(m)}$,
(4) $\operatorname{ind}_g(a^k)=k\cdot\operatorname{ind}_g(a) (\bmod \phi(m))$.

Fact: Let m be a positive integer. Then the only solutions to the congruence $x^2\equiv 1(\bmod m)$ are $x\equiv \pm 1(\bmod m)$ iff m has a primitive root.

Example (Cubic Lemma): If p is of the form $3k+2$, then every integer is a cube mod p. If p is of the form $3k+1$, then exactly one-third of the integers are cubes.
Let $p=3k+2$. Then $a^{3k+1}\equiv 1 (\bmod p)$. Now 3 is relatively prime to $3k+1$, hence there exist m, n such that $m(3k+1)+3n=1$, therefore $a=a^{m(3k+1)+3n}\equiv (a^{3k+1})^m(a^n)^3\equiv (a^n)^3 (\bmod p)$. For $p=3k+1$, let a be a primitive element mod p, then the incongruent elements mod p are $a,a^2,a^3,\ldots,a^{3k-2},a^{3k-1},a^{3k}\equiv 1$, and exactly one-third of them are cubes. (More formally: if m has a primitive root and if $k\,|\,\phi(m)$. Then the congruence $x^k-1\equiv 0 (\bmod m)$ has k solutions. Hence we can conclude that the number of incongruent kth power residues of m is $\phi(m)/(k,\phi(m))$.

Example (IMO 1990): Find all positive integers n such that $\frac{2^n+1}{n^2}$ is an integer.

Clearly n must be odd. Assume $3^k \| n$, then $3^{2k} | n^2 | 2^n+1$, hence $2^n \equiv -1 (\bmod 3^{2k})$, implies $3^{2k-1} | n$ by previous argument. Hence $2k-1 \le k \Rightarrow k \le 1$. We note $n=3$ is a solution. Now if n has a prime factor >3 and let p be the least such one. Then $2^n \equiv -1 (\bmod p)$ and hence $2^{2n} \equiv 1 (\bmod p)$. If d is the order of 2 modulo p, then $d | 2n$. If d is odd, then $d | n$ and hence $2^n \equiv 1 (\bmod p)$, a contradiction, so d must be even, say $d = 2d_1 | 2n$, get $d_1 | n$. Also $d | p-1$, or $2d_1 | p-1 \Rightarrow d_1 \le \frac{p-1}{2} < p$. But $d_1 | n$, so $d_1 = 1$ or 3. (p is the least prime of n other than 3). If $d_1 = 1$, then $d = 2$ and $2^2 \equiv 1 (\bmod p)$, a contradiction. If $d_1 = 3$, then $d = 6$ and $2^6 \equiv 1 (\bmod p)$, or $p | 63 \Rightarrow p = 7$. However the order of 2 modulo 7 is 3, which is odd, again a contradiction. Hence none of the other primes work.

Fact: Every prime has a primitive root. (Legendre)

Suppose p is prime. Let u be the smallest positive integer such that $a^u \equiv 1 (\bmod p)$ for every a relatively prime to p (u is the *least universal exponent* for p). Let g of an integer with order u modulo p. Then every integer relatively prime to p is a solution of $x^u \equiv 1 (\bmod p)$, so the congruence has $p-1$ solutions. By the Lagrange's theorem (non-congruence solutions of a polynomial with respect to its degree), the congruence has no more than u solutions, thus $u = p-1$. This means g is a primitive root.

Note (Lagrange's theorem): p is prime and $f(x)$ a polynomial of degree n not all coefficients divisible by p. Then the congruence $f(x) \equiv 0 (\bmod p)$ cannot have more than n incongruent solutions. Proof is by induction, true for $n = 0,1$ say. Let the theorem true for all polynomials of degree $n-1$, say. Now let $f(x)$ be of degree n, either it has no solution, or $f(a) \equiv 0 (\bmod p)$, then $f(x) = (x-a)q(x)$ (division algorithm). If b is another incongruent solution, then $0 \equiv f(b) \equiv (b-a)q(b) (\bmod p)$, but $b-a \not\equiv 0 (\bmod p)$, hence $q(b) \equiv 0 (\bmod p)$. Clearly not all coefficients of $q(x)$ are divisible by p and by induction $q(x)$ has no more than $n-1$ incongruent solutions.

Fact (Gauss): n possesses primitive roots iff n = 2, 4, p^a, $2p^a$, where p an odd prime.

Exercise 2

1. (IMO 1960, 1) Find all three-digit numbers such that for each number the quotient of its division by 11 is the sum of the squares of its digits. (550, 803).
2. (IMO1968, 2) Find all positive integers x such that the product of its digits in the decimal expansion is $x^2-10x-22$. (12).
3. (IMO1968, 6) Find with proof the sum $\left[\frac{n+1}{2}\right]+\left[\frac{n+2}{2^2}\right]+\ldots+\left[\frac{n+2^k}{2^{k+1}}\right]+\ldots$. (Use $\left[x+\frac{1}{2}\right]=[2x]-[x]$ to get n).

4. (IMO 1971, 3) There exist infinitely pairs of relatively prime numbers in the sequence $2^n - 3, n = 2,3,...$. (Use induction, if $2^{n_1} - 3,..., 2^{n_k} - 3$ are relatively prime to other, then there exists n_{k+1} such that $2^{n_{k+1}} - 3$ is relatively prime to each of them in the sequence. Indeed if $p_1, p_2, ..., p_r$ are all odd prime divisors of the sequence. Then they are odd and $2^{p_i - 1} \equiv 1 \pmod{p_i}$, hence $2^{(p_i-1)...(p_r-1)} \equiv 1 \pmod{p_i}$ for all i. Choose $n_{k+1} = (p_1 - 1)(p_2 - 1)...(p_r - 1)$. Let p be a common divisor of $2^{n_i} - 3$ and $2^{n_{k+1}} - 3$. Proceed to prove it is a contradiction, hence $2^{n_{k+1}} - 3$ is actually relatively prime to all $2^{n_i} - 3$).

5. (IMO 1974, 3) $\sum_{k=0}^{n} \binom{2n+1}{2k+1} 2^{3k}$ is not divisible by 5. (Expand $(1 \pm 2\sqrt{2})^{2n+1}$).

6. (IMO1975, 4) Let A be the sum of digit of 4444^{4444} and B the sum of digits of A. Find the sum of digits of B. (estimate and mod 9).

7. (IMO1978, 1) Let m, n be natural numbers with $1 \le m < n$. In their decimal representations, the last three digits of 1978^m and 1978^n are equal. Find m and n so that $m + n$ is smallest. (As $1978^n - 1978^m = 1978^m(1978^{n-m} - 1) \equiv 0 \pmod{10^3}$, we see $m \ge 3$. Set $m = 3$ and try to minmise $1978^{n-m} \equiv 1 \pmod{5^3}$. By Euler's theorem get $n - m = 100$, and hence $m + n = 106$.)

8. (IMO1978, 3) The set of positive integers is the disjoint union of $\{f(n)\}$ and $\{g(n)\}$, with $f(1) < f(2) < ...$ and $g(1) < g(2) < ...$; also $g(n) = f(f(n)) + 1$ for all $n \ge 1$. Find $f(240)$. (Beatty's theorem or Zeckendorff Representations of integers.)

9. (IMO 1979,1) Let $\frac{p}{q} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + ... - \frac{1}{1318} + \frac{1}{1319}$. Show that $1979 \mid p$. (The sum is $\frac{1}{660} + \frac{1}{661} + ... + \frac{1}{1319}$. Sum head and tail to see result.)

10. (IMO 1981, 3) Find the maximum possible value of $m^2 + n^2$, where $1 \le m, n \le 1981$ and $(n^2 - mn + m^2)^2 = 1$. (Fermat's descent, $(1,0) \to (1,1) \to (2,1) \to (3,2) \to ...$ to see solutions are actually Fibonacci numbers. Finally get the answer $987^2 + 1597^2$.)

11. (IMO1983, 3) a, b and c are pairwise relatively prime positive integers. Show that $2abc - ab - bc - ca$ is the largest integer which cannot be represented in the form $xbc + yca + zab$, where x, y, z are non-negative integers. (A theorem of Frobenius).

12. (IMO 1991, 6) Given $a > 1$, construct a bounded sequence $\{x_n\}$ such that $|x_i - x_j||i - j|^a \ge 1$ for all i, j. (Try $\{n\sqrt{2}\}$, Liuoville's theorem).

13. (IMO1994, 4) Determine all pairs of positive integers (m, n) such that $\frac{n^3 + 1}{nm - 1}$ is an integer.

Quadratic Reciprocity

Let m be an integer greater than 1, and a is an integer relatively prime to m**.** If $x^2 \equiv a \pmod{m}$ has a solution, then a is a quadratic residue of m. Otherwise a is a *quadratic nonresidue* of m. Now let p be a prime, the Legendre symbol $\left(\frac{a}{p} \right)$ is of value 1 if a is a quadratic residue of p. Otherwise it is assigned the value -1.

Fact: Let p be an odd prime, a and b be integers relatively prime to p. Then

(a) $\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod p$, and

(b) $\left(\frac{a}{p}\right)\left(\frac{b}{p}\right) = \left(\frac{ab}{p}\right)$.

(a) If $x^2 \equiv a \pmod p$ has a solution, then $a^{\frac{p-1}{2}} \equiv x^{p-1} \equiv 1 \pmod p$ (Fermat's little theorem). If $x^2 \equiv a \pmod p$ has no solution, then for each $i, 1 \le i \le p-1$, there is a unique j $(j \ne i)$ such that $ij \equiv a$. Therefore all integers from 1 to $p-1$ can be arranged into $\frac{p-1}{2}$ such pairs. Taking their product, we get

$a^{\frac{p-1}{2}} \equiv 1\cdot 2\cdot 3\cdot ...(p-1) \equiv (p-1)! \equiv -1 \pmod p$, (Wilson).

(b) Use (a).

(Part (a) is the Euler's criterion).

For any integer a and any natural number n define the numerically least residue of a mod n as the integer a' for which $a \equiv a' \pmod n$ and $-\frac{n}{2} < a' \le \frac{n}{2}$.

Example: The least residue of $1,2,3,4$ (mod 5) are respectively $1,2,-2,-1$. The least residue of $1,2,3,4,5,6$ (mod 7) are respectively $1,2,3,-3,-2,-1$.

Let p be an odd prime and $(a,p)=1$. Let a_j be the numerically least residue of $aj \pmod p$ for $j = 1,2,..., \frac{p-1}{2}$. Then

Fact: (Gauss' Lemma). $\left(\frac{a}{p}\right) = (-1)^l$, where l is the number of $1 \le j \le \frac{p-1}{2}$ such that $aj < 0$.

Example: $(p=5)$ $a=1$, then $a1=1$ and $a2=2$, $l=0$. $a=2$, then $a1=2$ and $a2=-1$, $l=1$. $a=3\equiv -2$, then $a1=-2$ and $a2=-4\equiv 1$, $l=1$. $a=4\equiv -1$, then $a1=-1$ and $a2=-2$, $l=2$. $(p=7)$ $a=2$, then $a2=-3$ and $a3=6\equiv -1$, $l=2$. $a=3$, then $a2=6\equiv -1$ and $a3=2$, $l=1$.

(Proof of Gauss' Lemma) Check that $|a_j|$ are distinct numbers from 1 to $\frac{p-1}{2}$. But $a_j \equiv aj \pmod p$. Hence $a^{\frac{p-1}{2}}\left(\frac{p-1}{2}\right)! \equiv a_1 a_2 ... a_{\frac{p-1}{2}} = (-1)^l \left(\frac{p-1}{2}\right)!$ (note the appearance of l). Now apply the Euler's criterion.

Example: If p is an odd prime, then $\sum_{k=1}^{p-1}\left(\frac{k}{p}\right) = 0$.

The squares are $1^2, 2^2, ..., \left(\frac{p-1}{2}\right)^2$, and they are half of all the residues.

Example: $\left(\frac{-1}{p}\right) = \begin{cases} 1, & p \equiv 1 \pmod 4. \\ -1, & p \equiv 3 \pmod 4. \end{cases}$

Use Euler $(-1)^{\frac{p-1}{2}} = \left(\frac{-1}{p}\right)$. (Or: check that $x^2+1 \equiv 0 \pmod{4k+3}$ has no solution, for $p = 4k+1$, use $x = 1\cdot 2\cdot \ldots \cdot 2k$ and apply Wilson to get $x^2 \equiv -1 \pmod{4k+1}$ solvable.)

<u>Example:</u> If p is an odd prime, then $\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}$, i.e. it is equivalent to say that $\left(\frac{2}{p}\right) = \begin{cases} 1, & p \equiv 1,7 \pmod 8. \\ -1, & p \equiv 3,5 \pmod 8. \end{cases}$

$p \equiv 1,5 \pmod 8$, then

$$2^{\frac{p-1}{2}}\left(\frac{p-1}{2}\right)! \equiv 2\cdot 4\cdot 6\cdot \ldots \cdot (p-1)$$
$$\equiv 2\cdot 4\cdot \ldots \cdot \frac{p-1}{2}\cdot\left(-\frac{p-3}{2}\right)\cdot \ldots \cdot (-5)\cdot(-3)\cdot(-1)$$
$$\equiv (-1)^{\frac{p-1}{4}}\left(\frac{p-1}{2}\right)!.$$

By canceling out $\left(\frac{p-1}{2}\right)!$, we get $\left(\frac{2}{p}\right) \equiv 2^{\frac{p-1}{2}} \equiv (-1)^{\frac{p-1}{4}}$. This implies 2 is a residue or nonresidue according to $p \equiv 1,5 \pmod 8$.

Similarly $p \equiv 3,7 \pmod 8$, then

$$2^{\frac{p-1}{2}}\left(\frac{p-1}{2}\right)! \equiv 2\cdot 4\cdot 6\cdot \ldots \cdot (p-1)$$
$$\equiv 2\cdot 4\cdot \ldots \cdot \frac{p-3}{2}\cdot\left(-\frac{p-1}{2}\right)\cdot \ldots \cdot (-5)\cdot(-3)\cdot(-1)$$
$$\equiv (-1)^{\frac{p+1}{4}}\left(\frac{p-1}{2}\right)!.$$

Hence $\left(\frac{2}{p}\right) \equiv 2^{\frac{p-1}{2}} \equiv (-1)^{\frac{p+1}{4}}$. This implies 2 is a residue or nonresidue according to $p \equiv 7,3 \pmod 8$.

<u>Example:</u> n is an odd integer, then every divisor of $2^n - 1$ is of the form $8k \pm 1$.

Indeed let $n = 2m+1$. Then $2^n = 2^{2m+1} \equiv 2\cdot(2^m)^2 \equiv 1 \pmod p$, this implies $2 = \left(\frac{1}{2^m}\right)^2$, thus $\left(\frac{2}{p}\right) = 1$ and the result follows from the last observation.

<u>Fact: (Quadratic Reciprocity)</u> For distinct odd primes p and q,
$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}.$$

Hence in particular if p and q are odd primes, then $\left(\frac{p}{q}\right)=\left(\frac{q}{p}\right)$ unless both p and q are of the form $4m+3$, in which case we have $\left(\frac{p}{q}\right)=-\left(\frac{q}{p}\right)$.

(Sketch of Proof): From Euler's criterion, we have Gauss' lemma.

Now $ja = p\left[\frac{ja}{p}\right]+t_j$, $1\le j\le \frac{p-1}{2}$. Summing up from $j=1$ to $j=\frac{p-1}{2}$, we have $a\sum_{j=1}^{\frac{p-1}{2}} j = p\sum_{j=1}^{\frac{p-1}{2}}\left[\frac{ja}{p}\right]+\sum_{j=1}^{\frac{p-1}{2}} t_j = pT+\sum_{j=1}^{\frac{p-1}{2}} t_j$. Also

$$\sum_{j=1}^{\frac{p-1}{2}} t_j = s_1+s_2+\ldots+s_m+r_1+r_2+\ldots+r_l$$

$$s_1+s_2+\ldots+s_m+(p-r_1)+(p-r_2)+\ldots+(p-r_l)-lp+2(r_1+r_2+\ldots+r_l)$$

$$=\sum_{j=1}^{\frac{p-1}{2}} j-lp+2(r_1+r_2+\ldots+r_l).$$

Together with $\sum_{j=1}^{\frac{p-1}{2}} j=\frac{p^2-1}{8}$ and the previous equation, we have

$$\frac{p^2-1}{8}(a-1)=p(T-l)+2(r_1+r_2+\ldots+r_n).$$

Now if a is odd (i.e., $(a,2p)=1$, and $a-1$ even), we get

$$\left(\frac{a}{p}\right)=(-1)^T=(-1)^{\sum_{j=1}^{\frac{p-1}{2}}\left[\frac{ja}{p}\right]}=(-1)^l.$$

Here $s_1,s_2,\ldots,s_m$ are those ja with remainders between 1 and $\frac{p-1}{2}$ when divided by p, $r_1,r_2,\ldots,r_n$ are those with remainders greater than $\frac{p}{2}$.

Now if a is another odd prime q. We want to evaluate $\left(\frac{q}{p}\right)\left(\frac{p}{q}\right)$. But $\left(\frac{q}{p}\right)=(-1)^{\sum_{j=1}^{\frac{p-1}{2}}\left[\frac{qj}{p}\right]}$ and $\left(\frac{p}{q}\right)=(-1)^{\sum_{k=1}^{\frac{q-1}{2}}\left[\frac{pk}{q}\right]}$ Counting these is exactly like counting integral points within the box $1\le x\le\frac{p-1}{2}$ and $1\le y\le\frac{q-1}{2}$. According to $qx>py$ or $qx<py$, we separate the points $1\le x\le\frac{p-1}{2}$ and $1\le y\le\frac{q-1}{2}$ into sets S_1 and S_2, estimating the sizes of sets S_1 and S_2, we

get $\sum_{j=1}^{(p-1)/2}\left[\frac{jq}{p}\right]+\sum_{k=1}^{(q-1)/2}\left[\frac{kp}{q}\right]=\frac{p-1}{2}\cdot\frac{q-1}{2}$, (the number of integral points within the box) and hence the law of quadratic reciprocity.

Example: 5 is a quadratic residue of primes of the form $10m\pm 1$ and nonresidure of primes of the $10m\pm 3$.

We have $\left(\frac{5}{p}\right)=(-1)^{\frac{5-1}{2}\cdot\frac{p-1}{2}}\cdot\left(\frac{p}{5}\right)=(-1)^{p-1}\left(\frac{p}{5}\right)=\left(\frac{p}{5}\right)$.

Now 5 is a quadratic residues of primes of the form $5n+1$ and $5n+4$, i.e., primes of the form $10m\pm 1$, and nonresidues otherwise.

Example: Determine those odd primes such that 11 is a quadratic residue.

$\left(\frac{11}{p}\right)=(-1)^{\frac{11-1}{2}\cdot\frac{p-1}{2}}\cdot\left(\frac{p}{11}\right)=(-1)^{\frac{p-1}{2}}\left(\frac{p}{11}\right)$.

From direct calculations, we have $\left(\frac{p}{11}\right)=\begin{cases}1, & p\equiv 1,-2,3,4,5 \pmod{11}\\ -1, & p\equiv -1,2,-3,-4,-5 \pmod{11}\end{cases}$, and $(-1)^{\frac{p-1}{2}}=\begin{cases}1, & p\equiv 1 \pmod 4\\ -1, & p\equiv -1 \pmod 4\end{cases}$. Using the Chinese Remainder theorem and by solving the corresponding systems, get 11 a residue if $p\equiv \pm1,\pm5,\pm7,\pm9,\pm19 \pmod{44}$ and a nonresidue if $p\equiv \pm3,\pm21,\pm13,\pm15,\pm17 \pmod{44}$.

Example: Find p such that $x^2+3\equiv 0 \pmod p$ has a solution.

We need $\left(\frac{-3}{p}\right)=\left(\frac{-1}{p}\right)\left(\frac{3}{p}\right)=1$. Now $\left(\frac{3}{p}\right)\left(\frac{p}{3}\right)=(-1)^{\frac{p-1}{2}}=\left(\frac{-1}{p}\right)$. Thus in general $\left(\frac{-3}{p}\right)=\left(\frac{-1}{p}\right)\left(\frac{3}{p}\right)=\left(\frac{-1}{p}\right)^2\left(\frac{p}{3}\right)=\left(\frac{p}{3}\right)$. Now $\left(\frac{p}{3}\right)=1$ iff $p\equiv 1$ mod 3. But $p\not\equiv 4 \pmod 6$, we have $p\equiv 1 \pmod 6$ as answers.

Example: If $p=2^n+1, n\geq 2$, then $3^{\frac{p-1}{2}}+1$ is divisible by p.

Note that n must be even, say $n=2k$. Now $\left(\frac{3}{p}\right)\equiv 3^{\frac{p-1}{2}} \pmod p$. However $p\equiv 1 \pmod 4$, and $p=2^n+1=2^{2k}+1=4^k+1\equiv 2 \pmod 3$, hence $\left(\frac{p}{3}\right)=-1$. Using the quadratic reciprocity, get $\left(\frac{3}{p}\right)\left(\frac{p}{3}\right)=(-1)^{\frac{p-1}{2}}=1$, so $\left(\frac{3}{p}\right)=-1$, or $3^{\frac{p-1}{2}}+1\equiv 0 \pmod p$.

Example (Putnam 1954): Determine if there are positive integers x and y satisfying $x^2+3xy-2y^2=122$.

Multiplying the expression by 4 and by completing squares, we have $(2x+3y)^2-17y^2=488$. Hence it suffices to show that $z^2\equiv 488\equiv 12 \pmod{17}$ has no solution. But $\left(\frac{12}{17}\right)=\left(\frac{4}{17}\right)\left(\frac{3}{17}\right)=\left(\frac{3}{17}\right)=\left(\frac{2}{3}\right)=-1$.

Example: (i) There exist infinitely many primes of the form $3k-1$. (ii) There exist infinitely many primes of the form $3k+1$.

In the first case, consider $3p_1p_2...p_N-1$, where p_i's are all the primes of the required form. For (ii), let $p_1,p_2,...,p_N$ be all the primes of the form equal to 1 mod 3. Consider $M=4(p_1p_2...p_N)^2+3$. If q is a prime factor of M, then $q\neq 2,3$ and is odd, and q is not a member of the list, so we must have $q\equiv -1 \pmod 3$, (p_i's are all primes $\equiv 1 \pmod 3$). As q divides M, hence -3 is a quadratic residue mod q. But $\left(\frac{-1}{q}\right)=(-1)^{\frac{q-1}{2}}$ and $\left(\frac{3}{q}\right)\left(\frac{q}{3}\right)=(-1)^{\frac{q-1}{2}}$, hence $\left(\frac{-3}{q}\right)=\left(\frac{-1}{q}\right)\left(\frac{3}{q}\right)=\left(\frac{q}{3}\right)=\left(\frac{2}{3}\right)=-1$, contradicting the observation that -3 is a quadratic residue mod q.

Example: There exist infinitely many primes of the form $4k+1$.

Suppose $p_1,p_2,...,p_N$ are all of them. Consider $(2p_1p_2...p_N)^2+1$ and p a prime divisor of it, then $(2p_1p_2...p_N)^2\equiv -1 \pmod p$. But we see this implies p is of the form $4k+1$ not in the list.

Example: Determine if there are integers a and b satisfying $2b^2+3 \mid a^2-2$.
If b is even, then $2b^2+3\equiv 3 \pmod 8$, if b is odd, then $2b^2+3\equiv 5 \pmod 8$. Hence $2b^2+3$ has prime factors of the form $8k\pm 3$. For primes of this form $(-1)^{\frac{p^2-1}{8}}=-1$. This violate $a^2\equiv 2 \pmod p$.

Example: If $m>1$ and $n>1$ are integers of the same parity, then 3^n-1 is not divisible by 2^m-1.

If m is even, then $2^m=4^{m/2}\equiv 1 \pmod 3$, hence 2^m-1 is divisible by 3, but 3^n-1 is not. Now suppose both m and n are odd, then $2^m-1\equiv 7 \pmod{12}$, since 2^m-1 also divides 3^n-1. (Note 3 divides $2^{m-2}-2$, hence 12 divides 2^m-8.) Hence 2^m-1 has a prime factor p of the form $p\equiv 5,7 \pmod{12}$. p also divides 3^n-1, hence $3^{n+1}\equiv 3 \pmod p$ is a quadratic residue mod p.
We check that 3 is not a quadratic residue modulo primes of the forms $p\equiv \pm 5 \pmod{12}$. Indeed $\left(\frac{3}{p}\right)\left(\frac{p}{3}\right)=(-1)^{\frac{p-1}{2}}$. If $p\equiv 5 \pmod{12}$, then $\left(\frac{3}{p}\right)=\left(\frac{2}{3}\right)=-1$. If $p\equiv -5\equiv 7 \pmod{12}$, then $\left(\frac{3}{p}\right)=-\left(\frac{1}{3}\right)=-1$.

Fact: If q is a factor of $2^{2^t}+1,$ then q is of the form $2^{t+1}k+1.$

Indeed if q is a factor of $2^{2^t}+1,$ then $2^{2^t}\equiv -1 \pmod q$. Let d be the order of 2 mod q, then $d=2^{t+1}.$ On the other hand, by Fermat's little theorem, $2^{q-1}\equiv 1 \pmod q$. Hence $2^{t+1}\,|\,q-1.$

Fact: When $t\geq 2,$ then q is of the form $2^{t+2}h+1.$

Indeed $q=2^{t+1}k+1.$ When $t\geq 2,$ we show $k=2h.$ Now $q\equiv 1 \pmod 8$. By the law of quadratic reciprocity, $2^{\frac{q-1}{2}}\equiv 1 \pmod q$. Hence $\frac{q-1}{2}=k2^t,$ thus

$1\equiv 2^{\frac{q-1}{2}}\equiv 2^{k2^t}=(2^{2^t})^k\equiv(-1)^k \pmod q$. Therefore
$1\equiv(-1)^k \pmod q$, or k is even.

Example (Sum of Squares): Every prime of the form $4k+1$ can be expressed as a sum of two relatively primes squares.

Indeed we know we have s satisfying $s^2\equiv -1 \pmod p$ (quadratic reciprocity or more elementary considerations). Consider $sx-y,\ 0\leq x,y<\sqrt{p}.$ There are $\left[\sqrt{p}\right]+1$ choices for each of x, y and since $(\left[\sqrt{p}\right]+1)^2>\left(\sqrt{p}\right)^2=p,$ at least two of the values of $sx-y$ are congruent mod p, say s $sx_1-y_1\equiv sx_2-y_2 \pmod p$. Let $x=x_1-x_2$ and $y=y_1-y_2,$ clearly x and y are not both zeros since the pairs are distinct. Then we have $sx\equiv y \pmod p$ or $-x^2\equiv s^2x^2\equiv y^2 \pmod p$. Then x^2+y^2 is a multiple of p, but $0<x^2+y^2<2\left(\sqrt{p}\right)^2=2p,$ it follows that $x^2+y^2=p.$ To show that x, y relatively prime, consider $d\,|\,x$ and $d\,|\,y,$ then $d\,|\,x^2+y^2=p,$ or $d=1.$

Remarks: Indeed the representation is unique up to obvious rearrangement or change of signs. Moreover if we have a two sums of two squares, then using
$(a^2+b^2)(c^2+d^2)=(ac+bd)^2+(ad-bc)^2=(ac-bd)^2+(ad+bc)^2,$
we have their product a sum of two squares. Thus finally (Fermat; Euler 1749): let n be a positive integer. Then n is a sum of two squares if and only if every prime divisor of n of the form $4k+3$ appears to an even power in the prime factorization of n.

Arithmetic Functions

x a real number, $[x]$ = Integral part of x = largest integer less than or equal to x, i.e., $x-1<[x]\leq x$. $x-[x]$ = Fractional part of x = $\{x\}$, i.e., $0\leq\{x\}<1$.

Facts

(i) $[x]+[y]\leq[x+y]$.
(ii) n positive integer, then $[x+n]=[x]+n$, $[x/n]=[[x]/n]$.

(iii) $[-x]=\begin{cases}-[x]+1, & x \text{ noninteger}\\ -[x], & x \text{ integer}\end{cases}$

(iv) n positive integer, a real, then $[a] + [a + 1/n] + [a + 2/n] + \ldots + [a + (n-1)/n] = [na]$.

(v) $\{x + y\} \le \{x\} + \{y\}$.

p prime, $p^s|m$, $p^{s+1}\nmid m$ (doesn't divide), then write $p^s\|m$. Want to find s such that $p^s\|n!$.

In fact $[n/p]$ of the numbers $p, 2p, \ldots, [n/p]p$ divisible by p, $[n/p^2]$ of the numbers $p^2, 2p^2, \ldots, [n/p^2]p^2$ divisible by p^2, …, then

$$s=\sum_{j=1}^{\infty}[\frac{n}{p^j}] \text{ (finite sum) (Legendre).}$$

Estimate: $[n/p^j] \le n/p^j$, thus $s \le n(1/p + 1/p^2 + \ldots) = n/(p-1)$, then $s \le [n/(p-1)]$

Example: Write n as $d_o+d_1p+\ldots+d_rp^r$, with $0\le d_i<p$. Then $\left[\frac{n}{p}\right]=d_1+d_2p+\ldots+d_rp^{r-1}$, $\left[\frac{n}{p^2}\right]=d_2p+\ldots+d_rp^{r-2},\ldots,\left[\frac{n}{p^r}\right]=d_r$. Sum up and reshuffle to get $d_0+d_1(p-1)+\ldots+d_r(p^r-1)-d_0=n-(d_0+d_1+\ldots+d_r)=n-v_p(n)$, and we have the formula $\sum_{j=1}^{\infty}[\frac{n}{p^j}]=\frac{n-v_p(n)}{p-1}$. Hence $v_p(n)$ is the sum of digits when n is expanded p-adically.

Example: Find the largest k such that $3^k\|1000!$.

$[1000/3] = 333$, $[1000/3^2] = 111$, $[1000/3^3] =37$, $[1000/3^4] = 12$, $[1000/3^5] = 4$, $[1000/3^6] = 1$. $k=333+111+37+12+4+1=498$.

Example: $[x + y] \ge [x] + [y]$, then $[m/p^j] \ge [n/p^j] + [(m - n)/p^j]$, implies $C_n^m=\binom{m}{n}=\frac{m!}{n!(m-n)!}$ an integer.

$f:N\to R$ multiplicative if $f(mn) = f(m)f(n)$ for relatively prime m and n.

Facts:

(i) $f(1) = f(1.1) = f(1)f(1)$, thus $f(1) = 1$ if $f(1) \ne 0$.

(ii) $n=p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_s^{\alpha_s}$, $f(n)=f(p_1^{\alpha_1})f(p_2^{\alpha_2})\cdots f(p_s^{\alpha_s})$.

(iii) f multiplicative, $g(n)=\sum_{d|n}f(d)$, then g is multiplicative. In fact gcd $(m,n) = 1$, d divides m, e divides n, then de divides mn, and every factor of mn is of this form. Thus $g(mn)=\sum_{de|mn}f(de)=\sum_{d|m,e|n}f(d)f(e)=\sum_{d|m}f(d)\sum_{e|n}f(e)=g(m)g(n)$.

Here we introduce several multiplicative functions.

$\phi(n)$ = natural numbers less than n which are relatively prime to n = Euler's totient function.

Check: $\phi(1) = \phi(2) = 1$, $\phi(3) = \phi(4) = 2$, $\phi(p) = p - 1$ if p is prime. How about $\phi(p^k)$? In fact, numbers less than p^k, but not relatively prime to p^k, are p, $2p$, …, $(p^{k-1}-1)p$, thus

$$\phi(p^k) = (p^k - 1) - (p^{k-1} - 1) = p^{k-1}(p-1) = p^k(1 - \frac{1}{p}).$$

Assume $\phi(n)$ is multiplicative, $n = p_1^{\alpha_1} \cdots p_s^{\alpha_s}$, then

$$\phi(n) = \phi(p_1^{\alpha_1}) \cdots \phi(p_s^{\alpha_s}) = p_1^{\alpha_1}(1 - \frac{1}{p_1}) \cdots p_s(1 - \frac{1}{p_s}) = n(1 - \frac{1}{p_1}) \cdots (1 - \frac{1}{p_s}).$$

Also we have $\sum_{d|n} \phi(d) = n$. In fact it suffices to check the case $n = p^k$ since $\sum_{d|n} \phi(d)$ is multiplicative. Suppose now $n = p^k$, then its divisors $d = 1, p, p^2, \ldots, p^k$, and $\sum_{d|p^k} \phi(d) = 1 + (p-1) + (p^2 - p) + \cdots + (p^k - p^{k-1}) = p^k$. OK.

<u>Mobius Function:</u> μ: $\boldsymbol{N} \to \boldsymbol{R}$ defined by $\mu(1) = 1$, $\mu(n) = 0$ if n contains a square factor, and $\mu(p_1 p_2 \cdots p_k) = (-1)^k$ if the p_i's are distinct prime factors. Check that this is a multiplicative function. Now $\nu(n) = \sum_{d|n} \mu(d)$ is multiplicative, and $\nu(p^k) = \sum_{d|p^k} \mu(d) = 1 + (-1) = 0$. Thus

$$\nu(n) = \begin{cases} 1, n = 1 \\ 0, n > 1. \end{cases}$$

Then we get the <u>Mobius Inversion Formula:</u> f an arithmetic function, $g(n) = \sum_{d|n} f(d)$, then

$$f(n) = \sum_{d|n} \mu(d) g(\frac{n}{d}).$$

<u>Proof:</u>

$\sum_{d|n} \mu(d) g(\frac{n}{d}) = \sum_{d|n} \mu(d) \sum_{e|\frac{n}{d}} f(e) = \sum_{e|n} f(e) \sum_{d|\frac{n}{e}} \mu(d) = \sum_{e|n} f(e) \nu(\frac{n}{e}) = f(n)$, since $\nu(n/e) = 1$ when $n = e$, and 0 otherwise.

Other multiplicative functions: $\tau(n)$ = number of divisors of n, $\sigma(n)$ = sum of divisors of n. To check that they are multiplicative, observe $\sigma(n) = \sum_{d|n} d$ and the identity function is multiplicative.

Also $\tau(mn) = \sum_{de|mn} 1 = \sum_{d|m} 1 \sum_{e|n} 1 = \tau(m)\tau(n)$.

Check that because the divisors of p^k are $1, p, p^2, \ldots, p^k$, we have $\tau(p^k) = k + 1, \sigma(p^k) = 1 + p + p^2 + \cdots + p^k = \frac{p^k - 1}{p - 1}$. And the general formulas may be obtained.

n is <u>perfect</u> if $\sigma(n) = 2n$. Examples: $n = 6, 28$.

<u>Theorem:</u> an even number n is perfect if and only if n is of the form $2^{p-1}(2^p - 1)$, where p and $2^p - 1$ are primes.

(Try $p = 2$, get 6, $p = 3$, get 28, $p = 5$, get 496, $p = 7$, get 8128.)

$n = 2^{p-1}(2^p - 1)$. Then factors of $n = 1, 2, \ldots, 2^{p-1}, (2^p - 1), 2(2^p - 1), \ldots, 2^{p-1}(2^p - 1)$. Sum up $\sigma(n) = (1 + 2 + \ldots + 2^{p-1}) + (2^p - 1)(1 + 2 + \ldots + 2^{p-1}) = 2^p(2^p - 1) = 2n$. Conversely, let $n = 2^k m$, m odd. Suppose $2^{k+1}m = 2n = \sigma(n) = \sigma(2^k)\,\sigma(m) = (2^{k+1} - 1)\,\sigma(m)$, implies $\sigma(m) = (2^{k+1}m)/(2^{k+1} - 1) = (2^{k+1} - 1 + 1)m/(2^{k+1} - 1) = m + m/(2^{k+1} - 1)$. Since m and $\sigma(n)$ are integers, we have $F = m/(2^{k+1} - 1) = 1$. (Note F is a factor of m, and if $F > 1$, then $\sigma(m) \geq 1 + m + F > m + F$.) This implies $m = 2^{k+1} - 1$ is prime and $k + 1 = p$.

The Riemann Zeta Function $\tau(s) = \sum_{n=1}^{\infty} \frac{1}{n^s},\ s = \sigma + it,$ with $\sigma > 1$.

Theorem: $\tau(s) = \prod_p (1 - \frac{1}{p^s})^{-1}, \sigma > 1.$

(In essence, this is the unique factorization of integers)

Proof: N a fixed positive integer, then

$\prod_{p \leq N} \frac{1}{(1 - \frac{1}{p^s})} = \prod_{p \leq N} (1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \cdots) = \sum_m \frac{1}{m^s},$ where m are positive integers with prime factors less than or equal to N.

But $\left| \sum_m \frac{1}{m^s} - \sum_{n \leq N} \frac{1}{n^s} \right| \leq \left| \sum_{n > N} \frac{1}{n^s} \right| \leq \sum_{n > N} \left| \frac{1}{n^s} \right| = \sum_{n > N} \frac{1}{n^\sigma} \to 0,$ as $N \to \infty.$

(What is left is $1/n^s$ with $n > N$ and prime factors of n less than or equal to N.

(i) $s \to 1,$ implies $\sum \frac{1}{n}$ diverges, infinitely many primes.

(ii) $x \leq ½$, then $x > -(\log(1-x))/2$, then $1/p > -(\log(1 - 1/p))/2$. Hence
$\sum_p \frac{1}{p} > \frac{1}{2} \sum_p \log(1 - \frac{1}{p})^{-1} = \frac{1}{2} \log \prod_p (1 - \frac{1}{p})^{-1},$ thus $\sum_p \frac{1}{p}$ diverges.

(iii) But $\sum_{p,q\ prime,\ p-q=2} \frac{1}{p}$ converges!

Exercises 3

1. Let m and n be integers greater than 1, show that $2m + 3n$ is divisible by 17 if and only if $9m + 5n$ is divisible by 17.

2. Suppose a^3 divides b^2, then a divides b. Is the conclusion true if a^2 divides b^3?

3. Show that for any positive integer n, there exists a multiple of n that consists of only 1's and 0's. Suppose further that n is relatively prime to 10, then there exists a multiple of n that consists only of 1's.
4. There exist infinitely many natural numbers a, such that $n^4 + a$ is composite (not prime) for every natural number n. (IMO1969 $(n^4 + 4k^2n^2 + 4k^4) - 4k^2n^2,$ complete square, etc).

5. There are $n - 1$ integers on a blackboard ranging from 2 to n. Two persons, A and B, take turn to rub off any one of them. Suppose A goes first. If the last two numbers left behind are relatively prime, then A wins, otherwise B wins. Determine who wins the game and find a winning strategy when n is (a) 2001 (b) 2006, (c) 2018.

6. Find all primes p such that $p + 10$ and $p + 14$ are primes.

7. Let $F_n = 2^{2^n} + 1.$ Show that if $m \neq n,$ then F_m and F_n are relatively prime.

8. Can you find n such that $\phi(n) = 14$?

9. When n is odd, $\phi(2n) = \phi(n),$ when n is even, $\phi(2n) = 2\phi(n).$

10. If 6 divides n, then $\phi(n) < \frac{1}{3}n.$

11. Let $n - 1$ and $n + 1$ be prime, $n > 4$, then $\phi(n) \leq \frac{1}{3}n.$

12. $\tau(n)$ is odd if and only if n is a perfect square.

13. $\sigma(n)$ is odd if and only of n is of the form $2^k m^2$, where m is odd.

14. If n is an integer > 1, then $\dfrac{2^n - 1}{n}$ is not an integer.

15. For every positive integer k, there exists an integer n with precisely k distinct prime factors such that n divides $2^n + 1$. (IMO2000).

16. $n > 1$, and n^2 divides $2^n + 1$, then $n = 3$. (IMO1990)

17. For any prime p, there exist infinitely many n such that $2^n - n$ is divisible by p. (Note $2^{p-1} \equiv 1 (\text{mod } p),$ take $m \equiv -1 (\text{mod } p)$ and consider $2^{m(p-1)} - m(p-1).$)

18. Let $\{a_1, a_2, \cdots, a_{101}\}$ and $\{b_1, b_2, \cdots, b_{101}\}$ be a complete set of residues modulo 101. Can $\{a_1 b_1, a_2 b_2, \cdots, a_{101} b_{101}\}$ be a complete set of residues modulo 101? (No: Wilson).

19. Let n be an odd integer, show that the last two digits of $2^{2n}(2^{2n+1} - 1)$ is 28.

20. For all positive integers i, let S_i be the sum of the products of 1, 2, …, $p - 1$ taken i at a time, where p is an odd prime. Show that $S_1 \equiv S_2 \equiv \cdots \equiv S_{p-2} \equiv 0 \ (\text{mod } p).$

21. Find all positive integers x, y and z satisfying $8^x + 15^y = 17^z$.

22. (IMO2001, 4) Let n be an odd integer greater than 1, let $k_1, k_2, ..., k_n$ be given integers. For each of the $n!$ permutations $a = (a_1, a_2, ..., a_n)$ of 1, 2, …,n, let $S(a) = \sum_{i=1}^{n} k_i a_i.$ Show that there are two permutations b and $c, b \neq c,$ such that $n!$ is a divisor of $S(b) - S(c).$

23. (IMO2001, 6) Let a, b, c, d be integers with $a > b > c > d > 0.$ Suppose that $ac + bd = (b + d + a - c)(b + d - a + c).$ Show that $ab + cd$ is not prime.

24. (IMO2002, 3) Find all pair of integers m and n such that there exist infinitely many positive integers a such that $\dfrac{a^m+a-1}{a^n+a^2-1}$ is an integer. ($(m,n)=(5,3)$).

25. (IMO2002, 4) Let n be an integer greater than 1. The positive divisors of n are $1=d_1<d_2<...<d_k=n.$ Define $D=d_1d_2+d_2d_3+...+d_{k-1}d_k.$ Show that (a) $D<n^2.$ (b) Determine all n such that D is a divisor of $n^2.$ ((b), if and only if n is prime).

26. (IMO2003, 2) Find all pairs of integers (a,b) such that $\dfrac{a^2}{2ab^2-b^3+1}$ is a positive integer. ($(a,b)=(2k,1),(b/2,b),$ or $((b^4-b)/2,b)$.)

27. (IMO2003, 6) Let p be a prime number. Show that there exists a prime number q such that for every integer $n,$ n^p-n Is not divisible by $q.$(Choose q a prime divisor of $\dfrac{p^p-1}{p-1}$ such that $q\not\equiv 1 \pmod{p^2}$.)

28. (IMO2004, 6) A positive integer is *alternating* if every two consecutive digits in its decimal representation are of different parity. Find all positive integers *n* such that *n* has a multiple which is alternating. (If and only if *n* not divisible by 20).

29. (IMO2005, 2) Let $a_1, a_2,...$be as sequence of integers with infinitely many positive terms and infinitely many negative terms. Suppose for each positive integer $n,$ $a_1,a_2,...,a_n$ leave n different reminders on division by $n.$ Show that each sequence occurs exactly once in the sequence.

30. (IMO2005, 4). Given $a_n=2^n+3^n+6^n-1, \quad n=1,2,3,...$ Find all positive integers relatively prime to all a_n. (Only answer is 1. Show any prime $p\geq 5$ divides a_{p-2}.)

31. (IMO 2006, 4) Find all integer pairs (x,y) satisfying $1+2^x+2^{2x+1}=y^2.$ ($(0,2),(0,-2),(4,23),(4,-23)$).

32. (IMO2008, 3) Show that there exist infinitely many positive integers n such that n^2+1 has a prime divisor which is greater than $2n+\sqrt{2n}.$ (If $p\equiv 1 \pmod 8),$ then $x^2\equiv -1 \pmod p$ has two solutions in $[1,p-1]$ whose sum is $p.$ If n is the smaller solution then $p\,|\,n^2+1$ and $n\leq (p-1)/2.$ Show $p>2n+\sqrt{10n}.$)

33. (IMO2009, 1) Let n be a positive integer and let $a_1,a_2,...,a_k$ $(k\geq 2)$ be distinct integers in the set $\{1,2,...,n\}$ such that n divides $a_i(a_{i+1}-1)$ for $i=1,2,...,k-1.$ Show that n does not divide $a_k(a_1-1).$

34. (IMO2010, 3) Find all functions $g:\mathbb{N}\to\mathbb{N}$ such that $(g(m)+n)(m+g(n))$ is a perfect square for all $m,n.$ ($g(n)=n+c$).

35. (IMO 2012, 6) Find all positive integers n for which there exist non-negative integers $a_1,a_2,...,a_n$ such that $\dfrac{1}{2^{a_1}}+\dfrac{1}{2^{a_2}}+...+\dfrac{1}{2^{a_n}}=\dfrac{1}{3^{a_1}}+\dfrac{2}{3^{a_2}}+...+\dfrac{n}{3^{a_n}}=1.$ ($n\equiv 1,2 \pmod 4$).

36. (IMO2013, 1) For any pair of positive integers k and $n,$ there exist k positive intgers $m_1,m_2,...,m_k$ (not necessarily different) such that $1+\dfrac{2^k-1}{n}=\left(1+\dfrac{1}{m_1}\right)\left(1+\dfrac{1}{m_2}\right)...\left(1+\dfrac{1}{m_k}\right).$ (Sort of induction on k).

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